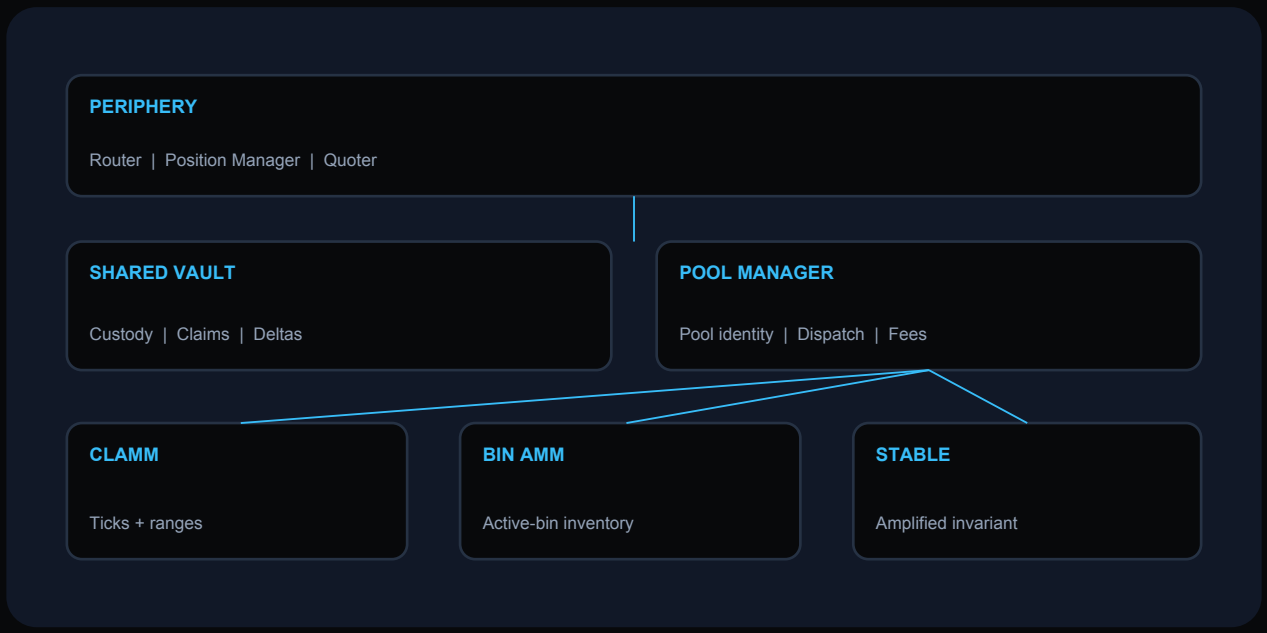
SUPADAO DEX / TECHNICAL PUBLICATION

# Protocol & Architecture

Shared settlement, specialized curve engines, explicit trust boundaries




VERSION 1.0.0 · 15 OCTOBER 2026 · UNDER TECHNICAL REVIEW

Repository-derived documentation. No independent audit or public deployment was confirmed.

# Document control

This publication is reproducible from Markdown sources in the SupaDAO DEX repository. It describes inspected source and is not legal, tax or investment advice.

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# Purpose

This guide isolates the technical architecture from the broader white paper. SupaDAO DEX is coordinated by a shared pool manager and vault while curve-specific engines retain their own state and mathematics.

# Topology

`SupaPoolManager` is the common entry point for pool initialization, liquidity changes, swaps, donations and protocol fees. `SupaVault` custodies assets, maintains ERC-6909 claims and enforces transient settlement. The manager selects `CLAMMEngine`, `BinAMMEngine` or `StableAMMEngine` based on `PoolKey.curveType`.

# Pool identity

A pool ID hashes ordered currencies, fee tier, tick/bin spacing, plugin address and curve type: `PoolId = keccak256(abi.encode(PoolKey))` Two pools sharing currencies are still independent if any field differs. The singleton architecture does not automatically combine their depth.

# Execution and accounting

Periphery contracts open a vault unlock callback. Each engine returns a packed signed token delta. The manager converts that result into transient obligations. Input transfer, output withdrawal and claims operations must clear the obligations before the callback ends (EIP-1153 net zero delta).

# Curve-specific design

• CLAMM: Q64.96 prices, initialized ticks, bitmap search, range positions and inside fee growth.

• Bin AMM: Active-bin inventory, discrete bin price and per-bin fee growth. Cross-bin traversal is not implemented in current source.

• Stable AMM: Two reserves, default amplification and iterative invariant solving. No decimal or rate normalization was found.

# Fees and plugins

Pool fee units are 1e-6 (hundredths of a bip). LP fees accrue through engine fee growth; protocol fees accrue by currency. Optional plugins receive lifecycle callbacks based on a permission bitmask. The dynamic-fee plugin is experimental and has a known stored-tick update limitation.

# Controls

Circuit breakers pause vault, pool or curve scopes. Guardians can pause; owner controls unpause. A timelock supports delayed operations and roles. The vault is UUPS upgradeable; the manager is not UUPS in the reviewed source.

# Integration and residual risk

Integrators must source canonical deployment addresses, reconstruct exact `PoolKey` values, quote current state and impose slippage/deadline checks. Primary risks are custom math, hooks, token behavior, administration, oracle limitations, depegs, liquidity gaps and MEV. No independent audit or public deployment was verified.